TypeScript Assignments – Interfaces & Types

Interfaces

1. Create an interface Student with name, age, and course.
2. Create an interface Product with id, name, and price.
3. Create an interface Employee with id, name, salary, and department.
4. Create an interface Car and create two car objects using that interface.
5. Create an interface BankAccount and write a function that accepts a BankAccount object.

Interface Extension

6. Create an interface Person with name and age. Extend it using an Employee interface.
7. Create a Vehicle interface and extend it using a Car interface.
8. Create a User interface and extend it with an Admin interface by adding an isAdmin property.
9. Create Student and CollegeStudent interfaces where CollegeStudent extends Student.
10. Create a Person interface, then create Employee and Manager interfaces that extend it. Create an object using the Manager interface.

Interface Merging

11. Create an interface Student with a name property. Declare the same interface again with an age property. Create an object containing both.
12. Create an interface Employee with id and declare it again with salary. Create an employee object.
13. Create an interface Product with name, then merge it with another declaration containing price.
14. Create two declarations of an interface User, adding different properties in each declaration.
15. Create a merged interface containing at least five properties across two interface declarations and create an object using it.

Type Aliases

16. Create a type alias Student for an object containing name, age, and course.
17. Create a type alias ID that can store either a number or a string.
18. Create a type alias Product and use it to create two product objects.
19. Create a type alias Employee and write a function that accepts an employee object.
20. Create a type alias for a function that accepts two numbers and returns a number. Create a function using that type.

Intersection Types

21. Create two types, Person and Employee, and combine them using an intersection type.

22. Create Address and Contact types and create a new type containing properties from both.
23. Create User and Admin types and combine them using &.
24. Create Product and Discount types and create a type that contains properties from both.
25. Create three types—Person, Employee, and Address—and create an intersection type that requires properties from all three. Create a valid object using that type.